DEPARTMENT OF ELECTRICAL ENGINEERING & ELECTRONICS

Final Year Project Report

Wireless Health Monitoring System

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April 2026

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*Software applications include but are not limited to, ChatGPT, Bing Chat, DALL.E, Bard

Acknowledgement

I would like to express my gratitude to my project supervisor for her continuous guidance, constructive feedback, and support throughout the development of this project. Her advice during bi-monthly meeting and encouragement were invaluable in shaping the direction of the work and ensuring successful project completion.I would also like to extend my gratitude to the project assessor for suggesting important points to include in the final report and identifying sections that could be further improved.

I would like to thank the laboratory team for prompt fabrication of 3D print and printed circuit boards and providing equipment during the hardware development of the project.

Special thanks are extended to my family and friends for their helpful suggestions during this project. Their input contributed to improving both the technical design and overall presentation of the work.

Abstract

The Final Year Project presents the development of a compact wireless health monitoring system capable of measuring physiological parameters within a single device and transmitting the collected data wirelessly to a secondary monitoring station for remote observation. The system consists of separate transmitter and receiver units, with controlled access. The developed prototype achieved an overall measurement accuracy of approximately ±5% while maintaining a low manufacturing cost of £175. A custom PCB HAT was designed to decrease jumper wire usage by 27%, along with a modular 3D-printed enclosure for improved portability. The proposed system offers potential to improve patient triage efficiency in healthcare setting and provide a practical monitoring solution for home environments.

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Chapter 1: Introduction

1.1 Introduction

This report presents a Final Year Project completed over sixteen weeks across two semesters. The project focused on designing a wireless device capable of collecting physiological data and transmitting it to a separate monitoring unit. The system consists of two modular units that communicate wirelessly.

The prototype was developed to incorporate key medical features and relevant standards found in commercial healthcare products. It is intended for use in home environments and nursing stations.

1.2 Background and Application

This project was undertaken in response to increasing patient inflow and the growing need for efficient patient assessment systems. The number of patients reaching out to the hospital is rising due to unhealthy lifestyle in the recent decade. NHS[1] statistics indicate an annual increase of 77,000 A&[2]E attendances between March 2024 and March 2025 [1]. Approximately 75% of patients are seen within 4 hours, below the 95% target shown in Figure 1.

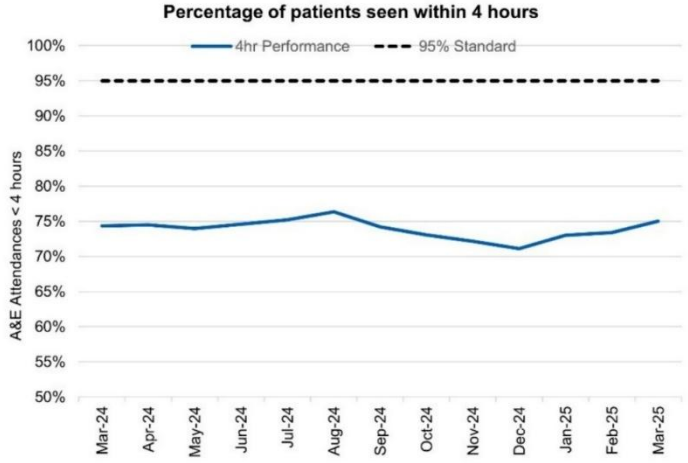


Figure 1:Percentage of A&E patients in England seen within four hours between March 2024 and March 2025

[1] National Health Service
[2] Accident and Emergency

Figure 2 illustrates that the patient journey in the emergency department consists of four stages: check-in, assessment, investigations, and outcome. Between the check-in and assessment stages, patients may be required to wait for at least two hours before triage is conducted. Once triage has been completed, the subsequent process is relatively quick, and treatment can then be provided.

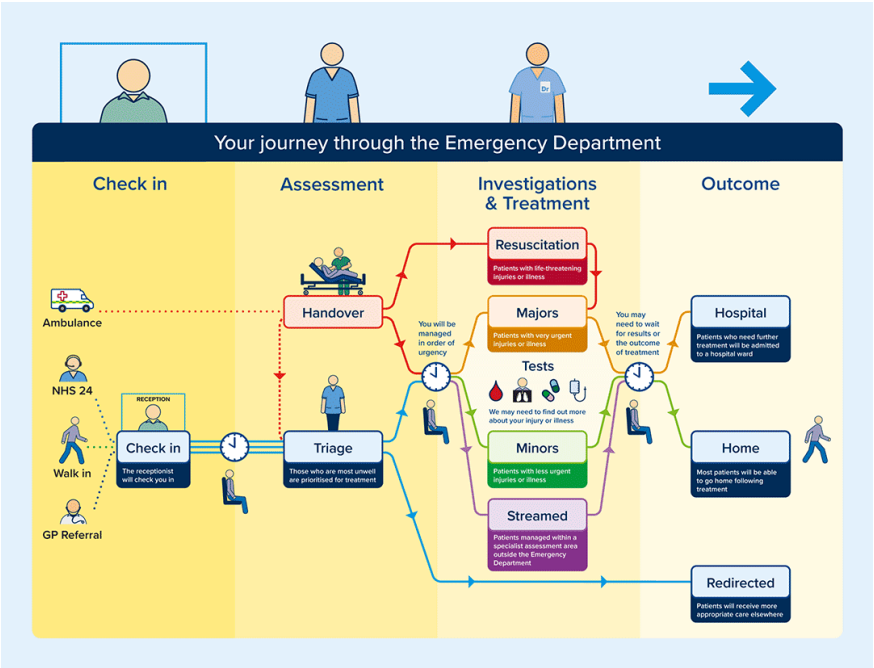


Figure 2: Emergency Department Journey

The report suggests a possible solution to improve the triage process efficiency and help increase the number of patients attended within four hours in A&E by developing a system that measures major vitals from one device.

1.3 State of the art solutions

Commercial vital sign monitoring systems include the IntelliVue VS30 from Philips. The system can measure blood pressure, blood oxygen saturation, pulse rate, and respiration rate, while also providing data logging and security features. However, such systems do not typically include integrated ECG measurement hardware and generally rely on dedicated hospital Wi-Fi networks for wireless data transmission. In addition, they are not specifically designed for portable device-to-device communication and are expensive. [2]

Consumer smartwatches such as the Garmin Fenix monitor several physiological parameters; however, they are not intended for medical diagnosis and are not suitable for continuous clinical monitoring [3]. These highlights demand for a cost-effective portable vital-sign monitor.

1.4 Project Objectives

The aim of the project is to develop a compact device capable of measuring major human vital signs simultaneously and transmitting the collected data for remote observation.

The objectives of the project are:

- To achieve vital sign measurement accuracy within ±5% of reference values
- To develop a prototype system costing less than £500.
- To design the product in consideration of relevant industry and medical standards.
- To design the system with a power consumption of less than 5 W and achieve a transmission range of up to 50 metres.

Chapter 2: Project Methodology

2.1 System Overview

The system is divided into two functional subunits: a sensing and transmission stage (transmitter unit), and a monitoring stage (receiver unit).

The system measures electrocardiogram (ECG) waveform, heart rate, blood oxygen saturation (SpO_2), and both ambient and body temperatures. The measured data is displayed locally and transmitted remotely through radio frequency, allowing doctors to view patients' vital signs from their offices. Security feature is incorporated at the monitoring stage to restrict access to unauthorised users.

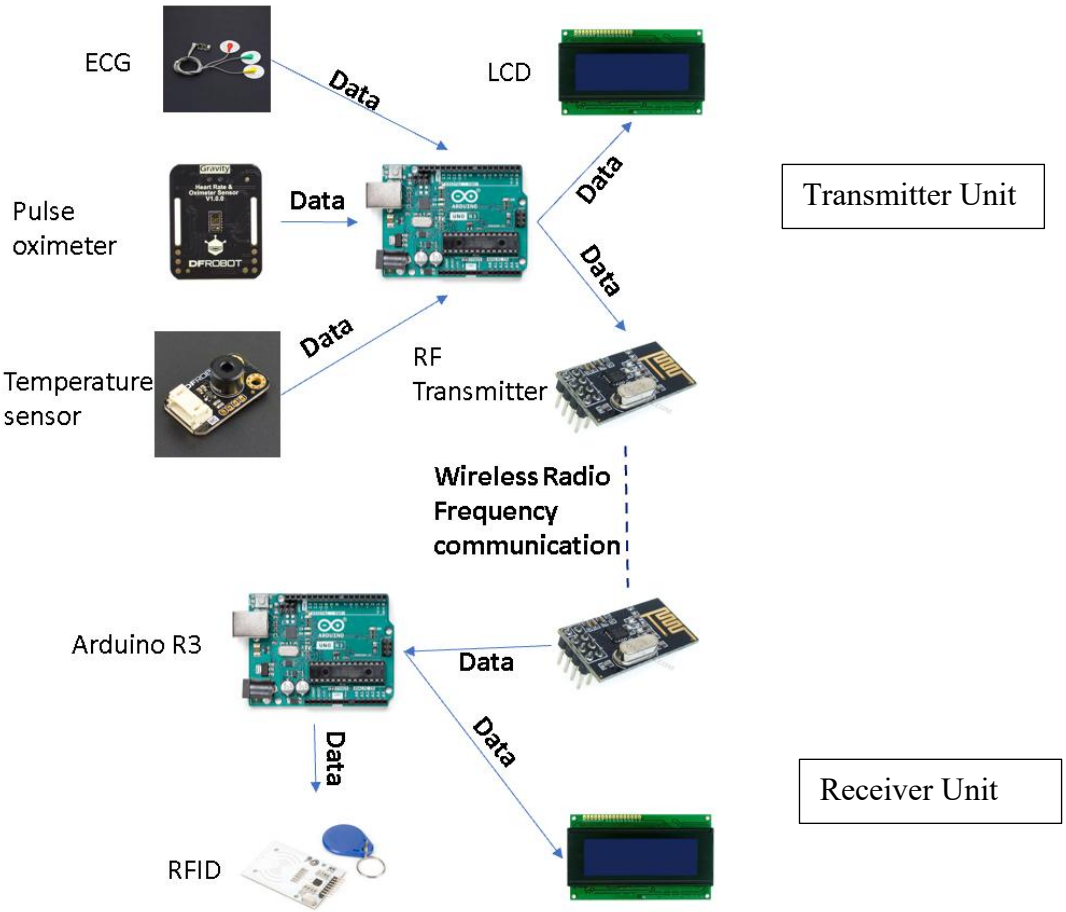


Figure 3:Overall system architecture of the wireless health monitoring system

Figure 3 illustrates the data flow within the system. The sensing hardware components transmit data to the microcontroller. The microcontroller then processes the acquired data and sends it wirelessly to the receiver unit through the radio frequency communication module. At the receiver side, the incoming data packet is unpacked by the microcontroller, and the data is displayed on the LCD once the user has been authorised by the RFID system.

In the system, components use two types of communication protocols: Serial Peripheral Interface (SPI) and Inter-Integrated Circuit (I^2C). The following hardware components use the I^2C communication protocol, operating at the clock frequency of 100 kHz:

I. LCD
II. Temperature sensor
III. Pulse oximeter

The hardware components utilising SPI communication protocol, operating at an 8 MHz clock frequency are:

I. Radio Frequency Trans receiver
II. RFID

The Electronic cardiogram module outputs electrical signal of the heart as an analogue signal to the microcontroller without requiring a communication protocol.
Within the discussion of the subunits of the system, the topics will be divided as follows:

I. The component list
II. Hardware schematic
III. Software program logic

2.2 Transmitter Unit Design

2.2.1 Material List- Transmitter Unit

Table 1:Transmitter Unit Components

Type of component	Component	Purpose
Power	Alkaline 9V Battery [5]	Power supply to the transmitter unit
Sensor	AD8232 Electronic Cardiogram [6]	Electrical activity of the heart
Sensor	MLX90614-DCC IR Temperature Sensor [4]	Body and ambient temperature measurement
Sensor	MAX30102 PPG Pulse oximeter [7]	Heart rate and blood oxygen saturation
Output	20x4 I2C LCD [8]	Display vital sign data
Communication module	nRF24L01 2.4 GHz Trans-receiver [9]	Radio frequency wireless communication

The transmitter subunit initially utilised an LM35CAZ temperature sensor IC instead of the MLX90614 temperature sensor. The LM35CAZ is a contact-based temperature sensor and is susceptible to ambient temperature interference. As a result, it is not ideal for medical environments due to hygiene concerns and reduced measurement accuracy, being approximately 40% less accurate than IR-based sensors according to the datasheet. [4]



Figure 5:LM35 Analogue Temperature Sensor, previously used



Figure 4:Non-contact Temperature Sensor, currently using

The pulse oximeter initially used was the Fermion series MAX30102. However, its 12-pin soldered connection and protruding underside pins made installation within the enclosure difficult. To improve reliability, it was replaced with the Gravity series MAX30102, which required only four jumper wire connections and no soldering. Its flat underside also allowed easier integration into the custom enclosure while providing sufficient clearance for the screw terminal connections

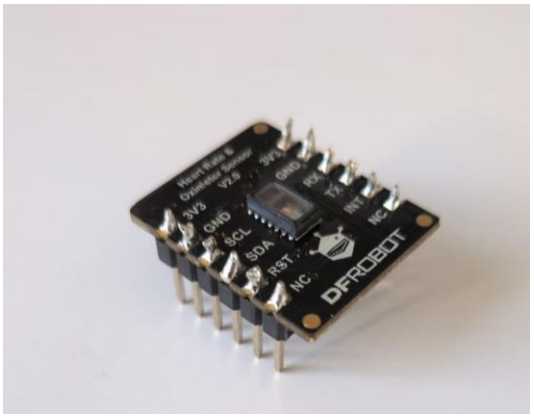


Figure 6:Fermion series: MAX30102, previously used

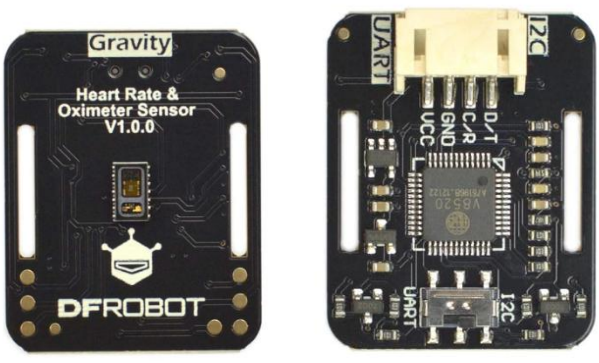


Figure 7: Gravity series: MAX30102, currently used

The LCD module was replaced from a 16×2 display utilising a parallel communication protocol to a 20×4 display with an I²C communication interface. This allows more information to be displayed simultaneously while reducing the number of connections by 25%.[8]

2.2.2 Schematic Diagram-Transmitter Unit

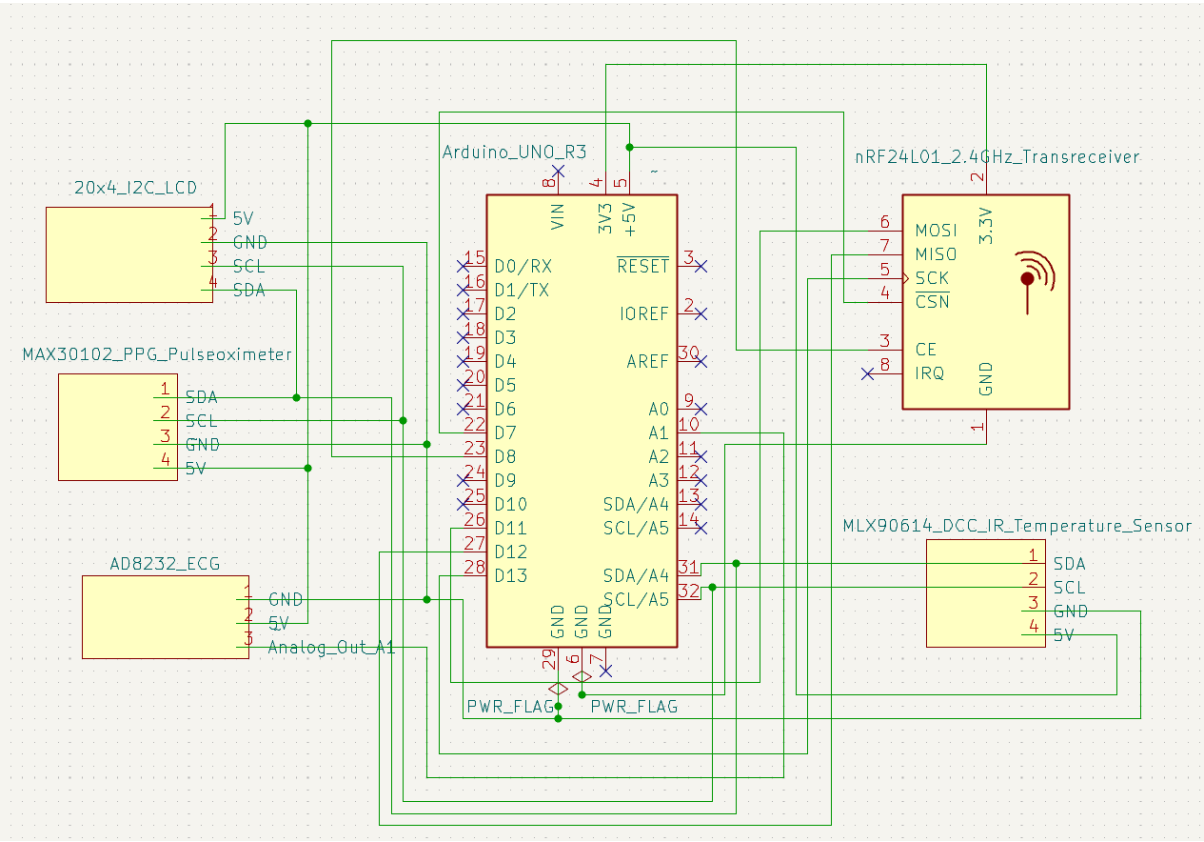


Figure 8:KiCAD Transmitter Schematic Diagram

The schematics were developed using the KiCAD application, Version 9.0.

The 20x4_I2C_LCD, MLX90614_DCC_IR_Temperature_Sensor, MAX30102_PPG_Pulseoximeter, and AD8232_ECG have custom symbols developed using the Symbol Editor. These symbols do not have specified dimensions; they are used to provide a visual representation of the connections and to aid in PCB[3] design. The Arduino_UNO_R3 and nRF24L01_2.4GHz_Transceiver symbols are imported from the built-in KiCad library.

[3] Printed Circuit Board

The PWR_FLAG and no-connect symbols are used to indicate a power net and intentionally unconnected pins, respectively, in the schematic, and are used for ERC[4]. ERC is a verification tool in KiCad that detects potential schematic issues such as unconnected pins, conflicting connections, and undriven power nets before fabrication of PCB.

2.2.3 Software Program Flow-Transmitter Unit

The system uses C++ language in Arduino IDE environment. The Arduino IDE version used for the project is 2.3.7.

Figure 9 summarises the program logic in the form of a flowchart. The transmitter unit is responsible for initialising the hardware sensors, collecting data, processing it through the microcontroller, and broadcasting data packets to the receiver through the communication module.

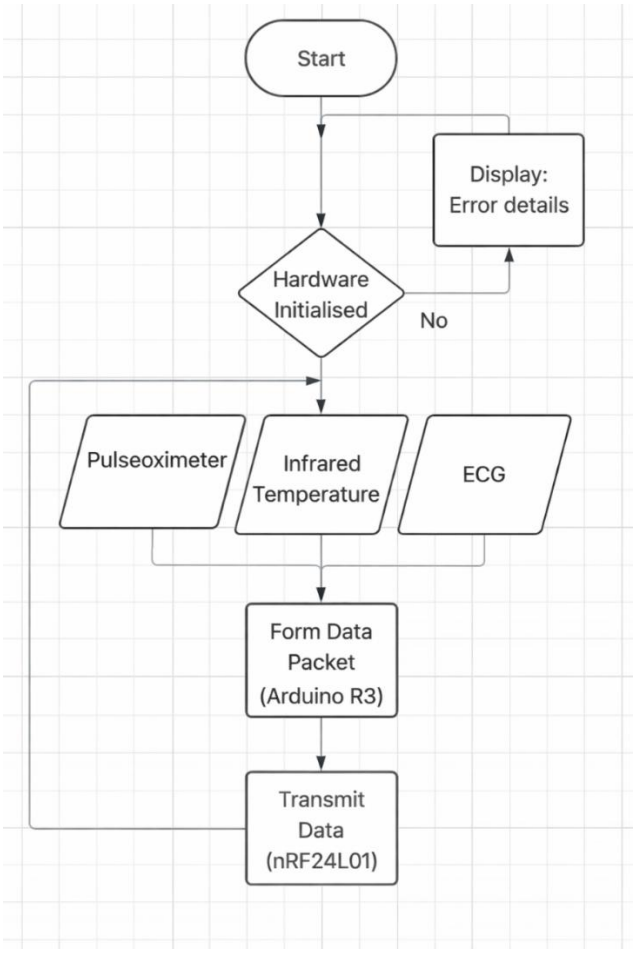


Figure 9:Transmitter Program Flowchart

[4] Electrical Rules Checker

The transmitter runs hardware connection test with IR Temperature and the pulse oximeter using the commands *while (NO_ERR != sensor.begin())* and *while (!MAX30102.begin())*. When the connections are misplaced, the while loop will return a false, suggesting faulty connection.

The communication address for the nRF24L01 module is “HLTH1” and is used to establish a communication link between the transmitter and receiver. The 2.4 GHz frequency band is widely used by multiple devices; therefore, the address ensures that data is received only by the intended receiver.

The temperature sensor measures the values in degrees Celsius. During development, a constant offset of 5.5 °C was observed at a measurement distance of 5 cm and was compensated through software calibration.

The ECG signal is sampled, and the analogue data is digitised using the 10-bit ADC of the microcontroller. The output waveform is displayed on the Arduino Plotter.

2.3 Receiver Unit Design

The receiver unit captures the data broadcast by the transmitter unit using the defined communication address. The packet is unpacked and displayed on the LCD after an authorised card is scanned.

2.3.1 Material List-Receiver Unit

Table 2:Receiver Unit Components

Type of component	Component	Purpose
Power	Alkaline 9V Battery	Power supply to the unit
Output	20x4 I2C LCD	Display vital sign data
Communication module	nRF24L01 2.4 GHz Trans-receiver	Radio frequency wireless communication
Security hardware	RFID WPI405 [10]	System security

2.3.2 Schematic Diagram-Receiver Unit

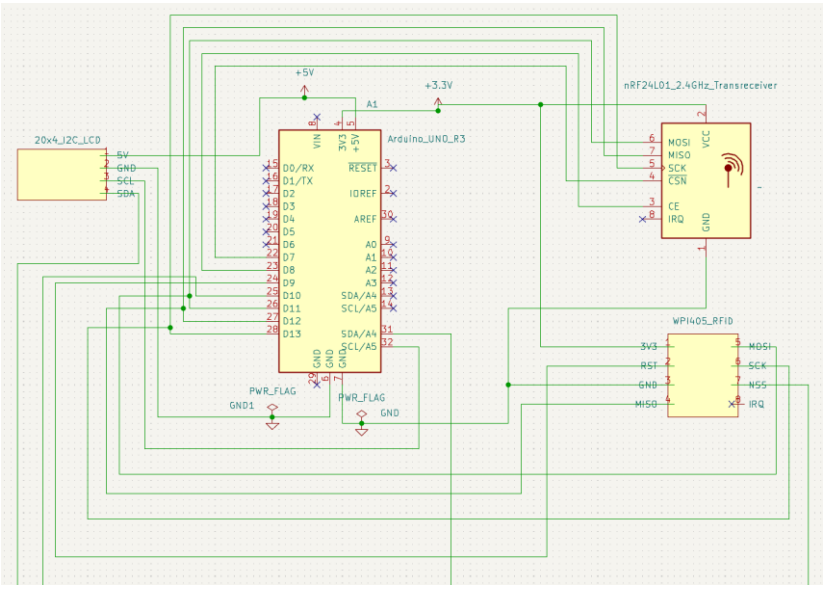


Figure 10:Receiver Unit Hardware Schematic

The 20x4_I2C_LCD and WPI405_RFID components use custom symbols developed using the Symbol Editor, while the Arduino_UNO_R3 and nRF24L01_2.4GHz_Transceiver symbols were imported from the built-in KiCad library.

The RFID module and the communication module operate using a 3.3 V power supply, while the LCD is powered by the 5 V output from the Arduino. The Arduino itself is powered by a 9 V battery and uses its built-in voltage regulator to step the supply voltage down to 5 V.

2.3.3 Software Program Workflow

The receiver unit begins by checking the status of its components and displaying any detected connection faults on the LCD. Once all modules are operating correctly, the system proceeds to user authentication.

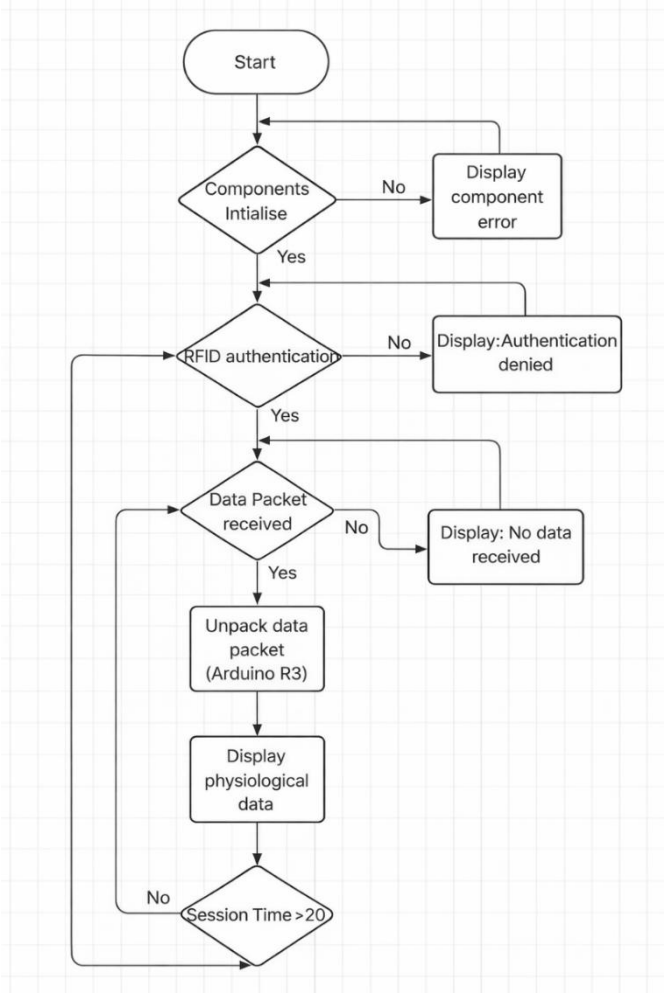


Figure 11:Receiver Unit Software Program Flow

During startup, the unit initialises the I2C and SPI communication protocols using *Wire.begin();* and *SPI.begin();* . The LCD is configured using *lcd.init()*; , the nRF24L01 radio module is started using radio.begin(); with the communication address set to “HLTH1”, and the RFID reader is initialised using *rfid.PCD_Init();* .

Access control is provided through RFID authentication. The user must tap a valid card whose UID matches one of the predefined authorised entries stored in the program. After successful verification, the physiological measurements are presented on the LCD.

The display session remains active for 20 seconds. Once this period has elapsed, the system returns to standby mode and requests authentication again before further data can be viewed.

Chapter 3: Product Design and Implementation

3.1 User Interface

3.1.1 System Start-up Screen

Transmitter Unit



Figure 12: Transmitter Unit Initialisation Screen

Receiver Unit

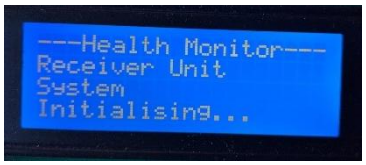


Figure 13: Receiver Unit Initialisation Screen

In the transmitter unit, a while loop is used to check the status of the pulse oximeter and infrared temperature sensor. The program displays "okay" when the sensors are healthy.

Otherwise, it displays "not okay" and continues the while loop until the sensors initialise successfully. Once all sensors are detected and functioning correctly, the system transitions to the physiological data display screen.

In the receiver unit, the system initialises the LCD, RFID, communication protocols and sets RF module into listening mode. Once the system has initialised successfully, the loading screen transitions into security screen requesting user to tap authorised card.

3.1.2 Receiver Unit Authentication Screen

Scan RFID Card



Authorised Access

Access Granted



Unauthorised Access

Access Denied

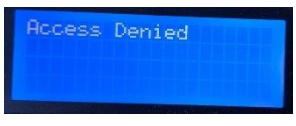


Figure 14: Receiver Unit Authentication Interface

Table 3:Summary of System Response for Authentication Interface

Card Type	System Response
Authorised Card	Displays “Access Granted” on the LCD and transitions into physiological data screen.
Unauthorised Card	Displays “Access Denied” on the LCD and requests to rescan authorised card

3.1.3 Physiological Data Display Screen

Transmitter Unit[5]

DS TE: < > s Itr:
ECG Status: Okay/ Fault
AT: < >C | BT: < >C
HR: < > BPM | SpO2: < > %



Figure 15:Transmitter Unit Physiological Data Screen

Receiver Unit

DR TE: < > s Itr:
ECG Status: Okay/ Fault
AT: < >C | BT: < >C
HR: < > BPM | SpO2: < > %



Figure 16:Receiver Unit Physiological Data Screen

Table 4:Abbreviation Table

Abbreviated Form	Full Form	Purpose
DR	Data Received	Indicates that data has been received from the transmitter
DS	Data Sent	Indicates that a data packet has been transmitted
TE	Time Elapsed	Displays the system uptime
Itr	Iterations	Displays the number of readings taken

The transmitter unit displays physiological measurements to two decimal places, while the receiver unit rounds the values to one decimal place. Figures 15 and 16 illustrate two different operating conditions in which the ECG signal is detected as healthy and faulty, respectively.

[5] < > Indicate a placeholder for values

3.2 PCB Design

The PCB was designed to reduce jumper wires and eliminate breadboard connections. Two KiCad layouts were created: transmitter and receiver PCBs.

The design approach was to implement an Arduino HAT configuration, where pin headers on the top side of the PCB to interface with external hardware components, while the bottom side is designed to mount directly onto the Arduino R3 microcontroller, as illustrated in the figures below.

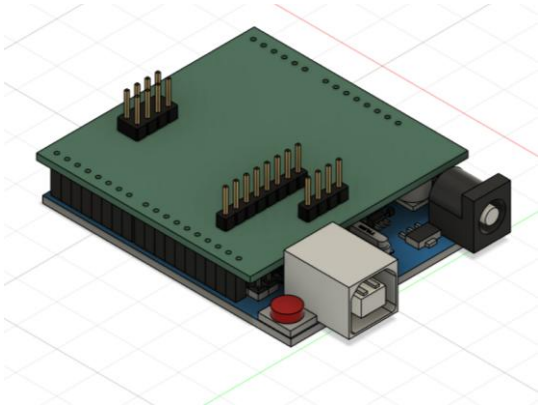


Figure 17:Isometric View 3D CAD Assembly of Receiver Unit

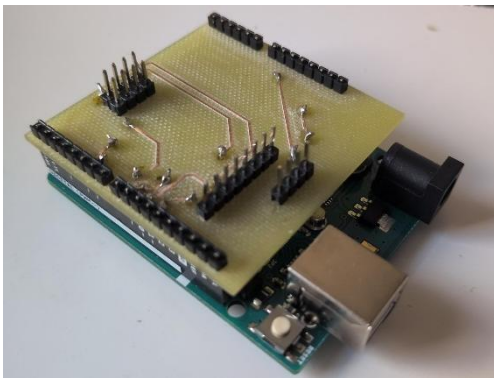


Figure 18:Isometric View of Arduino Hat-Receiver Unit

The pin headers utilise a standard 2.54 mm pitch and were manufactured using the university's PCB milling machine. The PCB was designed as a two-layer board consisting of Back Copper (B. Cu) and Front Copper (F. Cu), interconnected through vias.

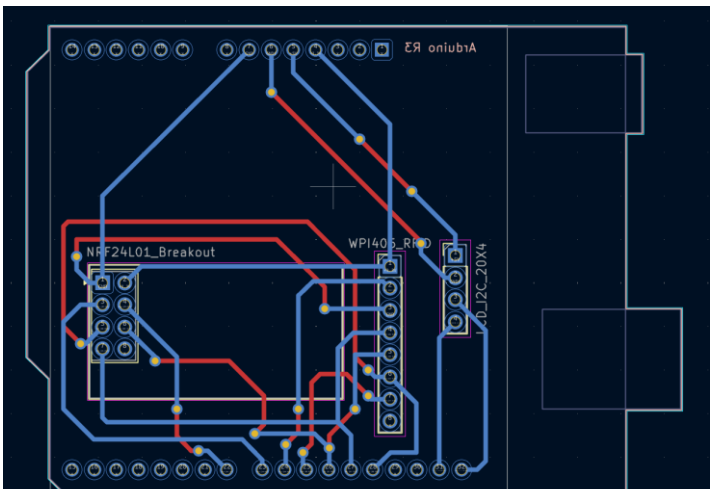


Figure 19:Receiver Unit Internal PCB connections, Blue-B. Cu, Red- F. Cu

Due to the limitation of the milling machine, which does not support plated through-hole (PTH) technology, all pin soldering was performed only on the Bottom Copper layer. Additionally, vias were implemented by inserting metal wires through the drilled holes and soldering them on both sides to establish electrical connectivity between the layers.

PCB connections were tested using continuity checks. During system integration, the CE pin of the communication module in the transmitter unit was found to be unresponsive. To isolate the source of the fault, each signal connection was bypassed individually by connecting the module directly to the Arduino, or through a breadboard, while testing the system step by step. This troubleshooting process identified the CE signal path as the cause of the issue. Continuity test confirmed that the solder joint of the CE pin on the PCB was weak.

3.3 3D Design

The 3D enclosure for the system was designed using Autodesk Fusion. The objective was a modular design allowing easy assembly of components. Each unit consists of three parts: a base enclosure, a component mounting slab, and a top lid.

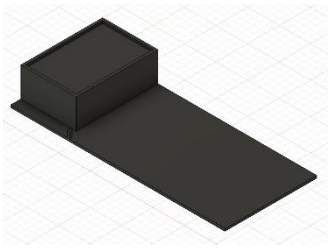


Figure 20:Receiver Unit Slit

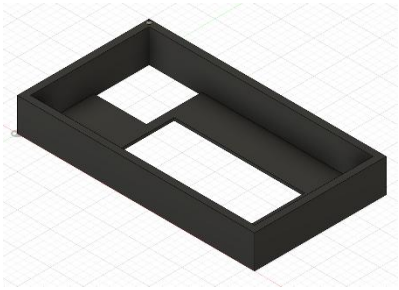


Figure 21:Receiver Unit Lid

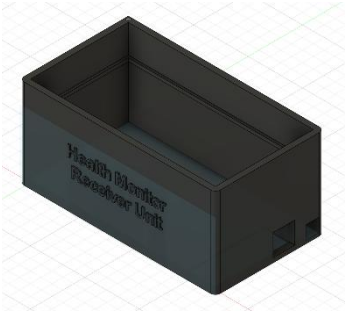


Figure 22:Receiver Unit Base Enclosure

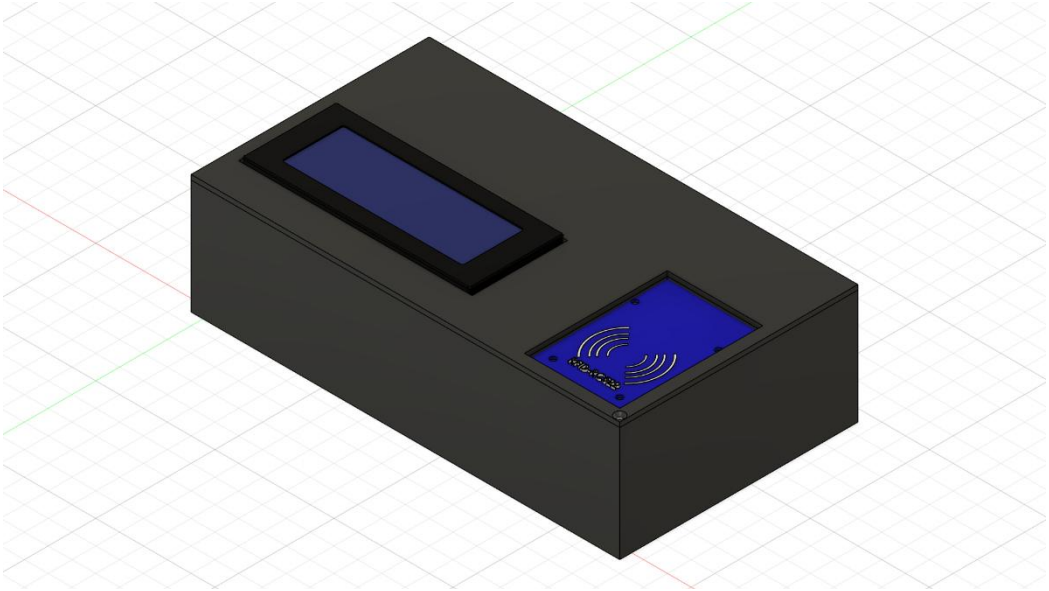


Figure 23: Receiver Unit Assembly

The base enclosure is designed to house the battery, microcontroller, RF transceiver, and includes designated openings for cable connections to a host computer as well as a power switch. The mounting slab is used to secure and expose externally interfacing components such as the LCD and IR sensors. The top lid serves as a protective cover and improves structural integrity of the unit.

A key feature of the enclosure design is that it does not require screws for assembly. Instead, a snap into place mechanism is used, allowing all parts to be securely joined while maintaining ease of disassembly for repairs.

Chapter 4: Results

The system was tested in the EEE 4th Laboratory to evaluate system accuracy. The results indicate that the system performed reliably in indoor conditions.

4.1 Electrocardiogram

The electrocardiogram (ECG) implemented in this system uses a three-lead configuration, which is lower complexity than a medical-grade twelve-lead ECG in terms of signal acquisition, spatial resolution and the number of electrodes required. Despite this, the system can measure basic cardiac electrical activity and identifying general rhythm irregularities such as bradycardia (slow resting heart rate), tachycardia (fast resting heart rate, and missing peaks.

The hospital-grade systems commonly operate at sampling rates of approximately 500 Hz[11], whereas the developed prototype samples at 200 Hz. Therefore, the system is intended for basic monitoring in triage process rather than clinical diagnosis. The ECG could not be compared with medical standard equipment due to ethical concerns but the provided matching results with heart rate when deducing heart rate from ECG waveform.

A sample ECG waveform representing normal heart activity, captured using the Arduino Serial Plotter, is shown in the figure below.

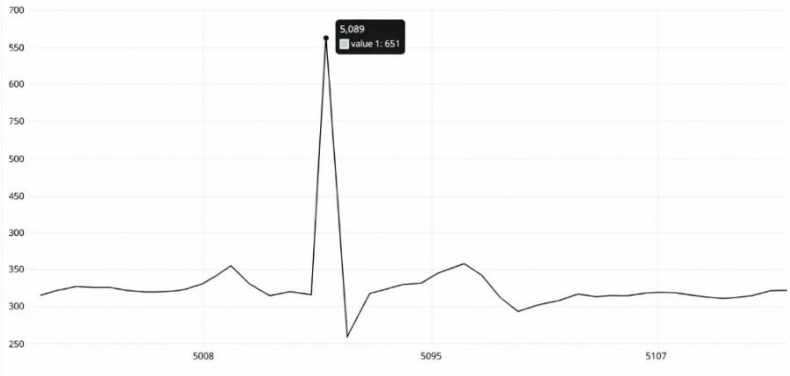


Figure 24:Electrocardiogram wave of healthy heart

4.2 Temperature Sensor

The temperature sensor demonstrated stable performance during testing, with readings within ±1 °C of values obtained from a domestic ambient thermometer and a medical thermometer under steady conditions. The manufacturer specifies an accuracy of ±0.5 °C over the temperature range of 0 °C to 50 °C [4]. Larger deviation observed in this project is due to measurement distance and limitations set by the software libraries.

The sensor can detect abnormal body temperatures. The table below summarises the ambient temperature results recorded under various conditions using a reference thermometer and the developed system. Extensive body temperature testing could not be conducted due to ethical constraints, and only limited temperature variation could be recorded.



Figure 25:Receiver unit displaying measured physiological data on the LCD alongside reference thermometers.

Table 5:Ambient Temperature Accuracy: Reference Thermometer Vs Prototype System

Reading	Condition	Reference (˚ C)	Prototype (˚ C)	Error (˚C)	Error (%)	Within ±5%
1	Direct Sunlight	28.8	29.9	1.1	3.82	Yes
2	In the Dark	22.8	23.5	0.7	3.07	Yes
3	Windy	16.4	17.2	0.8	4.88	Yes
4	Ambient	24.7	25	0.3	1.21	Yes
5	Near heat source (winter)	32.2	33.4	1.2	3.73	Yes
Mean Absolute Error						±0.82°C

4.3 Pulse Oximeter

The pulse oximeter did not consistently achieve the expected ±5% accuracy during initial testing. The sensor typically required around 10 seconds to stabilise, with noticeable heart rate fluctuations. These variations were mainly caused by the averaging technique used to combine multiple samples before display. After the averaging process was removed, the measurement accuracy improved to within ±5% in most of the test cases.

The pulse oximeter could not be tested in multiple scenarios, and the oxygen saturation did not have much variability as for a healthy human it is in between 95% to 100%[12]. The table below compares heart rate in three different scenarios and compares between reference pulse oximeter and pulse oximeter in the developed system.



Figure 26:Transmitter unit displaying measured physiological data on the LCD alongside reference pulse oximeter (Smartwatch).

Table 6:Pulse Oximeter Accuracy: Reference Smart Watch Vs Prototype System

Reading	Condition	Reference (BPM)	Prototype (BPM)	Error (˚C)	Error (%)	Within ±5%
1	Resting	75	78	3	4	Yes
2	Uphill walk	119	128	9	7.56	No
3	Focused Work	88	92	4	4.55	Yes
Mean Absolute Error						5.33BPM

The sensor also operates in reflectance mode, unlike transmission-based clinical pulse oximeters. The absence of an opposite detector reduces precision and increases sensitivity, which may further contribute to the observed variations.

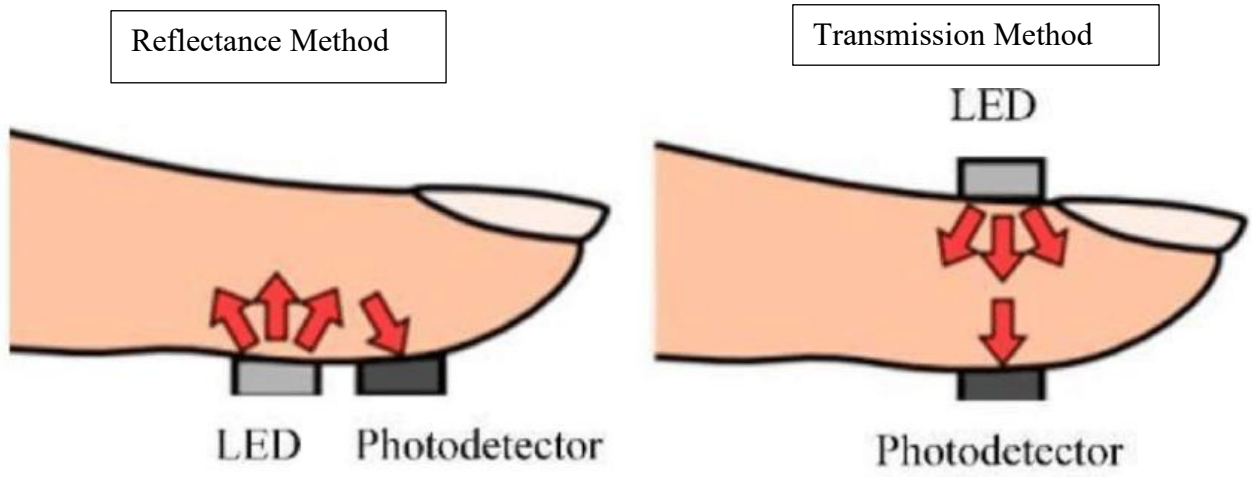


Figure 27:Pulse oximeter Sensing Method [11]

4.4 Radio frequency Trans-Receiver

The communication module transmitted data at a rate of 1 Mbps operating at 2.4 GHz was observed during testing. Each data packet of struct data type consists of 20 bytes; therefore, the transmission time is estimated to be around 160 microseconds.

$$t = \frac{Packet\ Size\ (bits)}{Data\ Rate} = \frac{20\ \times 8}{1 \times 10^6} = 160\mu s$$

The transceiver was able to maintain communication over approximately 50 metres under line-of-sight conditions in indoor environment.

This performance satisfies the project objective of enabling communication between units located in different areas, such as a nursing station and a doctor's office.



Figure 28:Successful wireless communication between transmitter and receiver units located on the third and fourth floor laboratories.

4.5 Current, Power and Battery Capacity

4.5.1 Transmitter Unit Power Consumption

The voltage requirements and current consumption of each component were obtained from their respective datasheets.

$$Power = Input\ Voltage\ (V) \times Current\ Drawn(I)$$

Device	Voltage Requirements (V)	Current Drawn (A)	Power (W)
Arduino Uno R3	5	50m	250m
nRF24L01	3.3	11m	36m
MAX30102	3.3	30m	100m
MLX90614	5	5m	25m
AD8232	5	0.2m	0.66m
LCD 20x4	5	30m	150m
Total	N/A	**126.2m**	**561m**

Table 7:Transmitter Unit Power Calculation Summary

4.5.2 Receiver Unit Power Consumption

Table 8:Receiver Unit Power Calculation Summary

Device	Voltage Requirement (V)	Current Drawn (A)	Power (V x I) (W)
Arduino Uno R3	5	50m	250m
nRF24L01	3.3	12m	39.6m
LCD	5	30 m	250m
MFRC522 RFID	3.3	20 m	66m
Total	N/A	**112 m**	**506m**

4.5.3 Battery Capacity

$$\text{Energy} = \frac{\text{Battery Capacity (Q) X Output Voltage (V)}}{1000}$$

9V battery's electrical capacity is 500mAh [5]

$$\text{Energy} = \frac{500\text{mAh X } 9\text{V}}{1000} = 4.5Wh$$

$$\text{Battery Duration} = \frac{\text{Battery Energy (Wh)}}{Power\ Consumed\ (W)}$$

Transmitter Battery

$$\text{Battery Duration} = \frac{4.5\text{Wh}}{0.561\ W} = 8\ hours$$

Receiver Battery

$$\text{Battery Duration} = \frac{4.5\ \text{Wh}}{0.506} = 9\ hours$$

The realistic battery duration will be around 60% of the rated battery duration of the system because the Arduino only requires 5V for supply and the battery provides 9V. The energy is wasted passing through the voltage regulator.

Chapter 5: Discussion and Conclusion

5.1 Ethical Considerations

The ethical concern associated with the project relates to the handling of patient data. To address this, the system was designed so that no personal information is attached to the transmitted data packets. Only physiological measurements are collected and displayed locally and remotely, without profiling the individual to whom the data belongs.

Although the radio frequency module broadcasts the measured data, the transmitted packets cannot be interpreted meaningfully without the matching communication address. In addition, access to the displayed data on the receiver unit is protected through the RFID authentication system. As a result, only authorised personnel can view patient measurements in real time.

System testing was limited to the author of this report to avoid potential safety risks to other users during evaluation stage.

5.2 Standards Compliance and Sustainability

The project was developed with consideration of international standards, with selected components being RoHS and CE compliant to support recognised environmental and safety requirements. Wherever possible, components were sourced locally to reduce transportation-related carbon emissions, and the PCB was manufactured using the university's in-house milling machine rather than being outsourced, thereby reducing shipping requirements.

The system was also designed in consideration to minimise waste. Individual components can be replaced easily without the use of screws, improving repairability, extending product lifespan, and enhancing the sustainability of the design.

5.3 Learning Experience and Future Work

The project provided a diverse skill development opportunity, ranging from C++ programming, hardware integration, and research work to PCB design and 3D modelling. It also helped develop key engineering skills relevant to industry, including awareness of medical product standards and ethical considerations.

In the future, the hardware could be upgraded with medical-grade sensors, secure patient data logging with encryption, and a local display for viewing the ECG waveform. The system could also be developed for IEC compliance.

5.4 Conclusion

The project objectives were achieved, with the developed system demonstrating effective overall performance. The temperature sensor operated within ±5% accuracy, while the pulse oximeter met this target for most measurements. Although the ECG waveform was not directly compared with a medical-grade reference system, it provided a reasonable representation of the heart's electrical activity for basic monitoring.

The £175 prototype considered IEC 60601-1 safety principles and used a 9V battery supply to reduce mains hazards. The combined power consumption of both units was approximately 1.1 W, providing a battery operating time of over 6 hours.

A key strength was successful integration of eight devices using multiple protocols with reliable operation. The custom PCB reduced wiring substantially while the enclosure reduced system footprint by 45.6cm^2.

Overall, the project delivered a low-cost prototype capable of measuring vital signs and transmitting data wirelessly for remote monitoring.

Chapter 6: References

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Appendix A : Pin Connections and Software Libraries

The PCB copper tracks route the Arduino pins to the corresponding components listed in the Connected Components column. During prototyping and testing, pins shared by multiple modules were interconnected using a breadboard.

Transmitter Pin Connections

Arduino Pin	**Connected Components**	**Signal Type**
A1	AD8232 ECG Output	Analogue Input
A4 (SDA)	MLX90614, MAX30102, LCD 20×4	I2C Data
A5 (SCL)	MLX90614, MAX30102, LCD 20×4	I2C Clock
D8	nRF24L01 (CE)	SPI Control
D7	nRF24L01 (CSN)	SPI Control
D11	nRF24L01 (MOSI)	SPI Data Line
D12	nRF24L01 (MISO)	SPI Data Line
D13	nRF24L01 (SCK)	SPI Clock
5V	LCD 20x4, MLX90614, AD8232, MAX30102	Power
3.3V	nRF24L01	Power
GND	All modules	Ground

Receiver Pin Connections

Arduino Pin	**Connected Components**	**Signal Type**
A4 (SDA)	MLX90614, MAX30102, LCD 20×4	I2C Data
A5 (SCL)	MLX90614, MAX30102, LCD 20×4	I2C Clock
D8	nRF24L01 (CE)	SPI Control
D9	MFRC522 (RST)	Reset
D10	MRFC522 SDA (SS)	SPI Chip Select
D7	nRF24L01, MFRC522 (CSN)	SPI Control
D11	nRF24L01, MFRC522 (MOSI)	SPI Data Line
D12	nRF24L01, MFRC522 (MISO)	SPI Data Line
D13	nRF24L01, MFRC522 (SCK)	SPI Clock
5V	LCD 20x4	Power
3.3V	nRF24L01, MFRC522	Power
GND	All modules	Ground

The software libraries and their functions are shown in the table below:

Library	**Purpose**
Wire.h	Enables I2C communication between the microcontroller and sensors such as the MLX90614 temperature sensor, MAX30102 SpO_2 sensor, and LCD display
LiquidCrystal_I2C.h	Controls the I2C LCD display used to show vital sign readings and system information
DFRobot_MLX90614.h	Interfaces with the MLX90614 infrared temperature sensor to measure non-contact body temperature
DFRobot_BloodOxygen_S.h	Interfaces with the MAX30102 sensor to measure heart rate and blood oxygen saturation (SpO_2)
SPI.h	Enables SPI communication protocol required for communication with the nRF24L01 wireless module
nRF24L01.h	Provides register definitions and configuration for the nRF24L01 radio frequency transceiver module
RF24.h	Handles wireless data transmission and reception between transmitter and receiver units using the nRF24L01 module
MFRC522.h	RFID reader interface for reading RFID card/tag data

Appendix B : Transmitter Unit Code

```
#include <Wire.h>                    // Library to enable I2C protocol
#include <LiquidCrystal_I2C.h>       // Library for LCD
#include <DFRobot_MLX90614.h>        // Library for IR sensor
#include <DFRobot_BloodOxygen_S.h>   // Library for MAX30102
#include <SPI.h>                     // Library for SPI communication protocol
#include <nRF24L01.h>                // Library for nRF24L01 module
#include <RF24.h>                    // Library to send and receive data

#define I2C_ADDRESS 0x57
DFRobot_BloodOxygen_S_I2C MAX30102(&Wire, I2C_ADDRESS);

RF24 radio(8, 7);
const byte address[6] = "HLTH1";

struct HealthPacket {
  int ecg;
  float ambientTemp;
  float bodyTemp;
  int bpm;
  int spo2;
  unsigned long timeSec;
  int iteration;
};

HealthPacket packet;

unsigned long startMillis;
unsigned long seconds;

DFRobot_MLX90614_I2C sensor;
LiquidCrystal_I2C lcd(0x20, 20, 4);

const int heartPin = A1;
int iteration = 0;

void setup() {

  Serial.begin(9600);
  Wire.begin();

  lcd.init();
  lcd.backlight();

  radio.begin();
```

```
  radio.openWritingPipe(address);
  radio.stopListening();

  Serial.println("\n---Initialising sensors---");

  startMillis = millis();
  seconds = 0;

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("---Health Monitor---");

  while (NO_ERR != sensor.begin()) {
    Serial.println("IR sensor (MLX90614) failed");
    delay(3000);
  }

  Serial.println("MLX90614 okay!");
  lcd.setCursor(0, 1);
  lcd.print("MLX90614 okay");

  while (!MAX30102.begin()) {
    Serial.println("MAX30102 Failed");
    delay(1000);
  }

  Serial.println("MAX30102 Okay!");
  MAX30102.sensorStartCollect();

  lcd.setCursor(0, 2);
  lcd.print("MAX30102 okay");

  delay(5000);
}

void loop() {

  iteration += 1;
  seconds = (millis() - startMillis) / 1000;

  int heartValue = analogRead(heartPin);

  float ambientTemp = sensor.getAmbientTempCelsius();
  float objectTemp = sensor.getObjectTempCelsius() + 5.5;

  /* ===== SINGLE PULSEOXIMETER READING (NO AVERAGING) ===== */

  MAX30102.getHeartbeatSPO2();
```

```
int spo2 = MAX30102._sHeartbeatSPO2.SPO2;
int bpm = MAX30102._sHeartbeatSPO2.Heartbeat - 40;

if (spo2 < 0) spo2 = 0;
if (bpm < 0) bpm = 0;

Serial.println("\n---------------------------");

Serial.println("Data Sent!");
Serial.print("Time Elapsed: ");
Serial.print(seconds);
Serial.print(" s | Iteration: ");
Serial.println(iteration);

Serial.print("ECG Value for Plotter: ");
Serial.println(heartValue);

Serial.print("Ambient temperature : ");
Serial.print(ambientTemp);
Serial.println(" °C");

Serial.print("Object temperature : ");
Serial.print(objectTemp);
Serial.println(" °C");

Serial.print("SPO2: ");
Serial.print(spo2);
Serial.println("%");

Serial.print("HeartRate: ");
Serial.print(bpm);
Serial.println(" BPM");

/* ================= LCD DISPLAY ================= */

lcd.clear();

lcd.setCursor(0, 0);
lcd.print("DS TE:");
lcd.print(seconds);
lcd.print("s Itr:");
lcd.print(iteration);

lcd.setCursor(0, 1);
lcd.print("ECG Status:");

if (heartValue > 10) {
```

```
    lcd.print("Okay");
  } else {
    lcd.print("Fault");
  }

  lcd.setCursor(0, 2);
  lcd.print("AT:");
  lcd.print(ambientTemp);
  lcd.print("C |BT:");
  lcd.print(objectTemp);
  lcd.print("C");

  lcd.setCursor(0, 3);
  lcd.print("HR:");

  if (bpm < 40) {
    lcd.print("-");
  } else {
    lcd.print(bpm);
    lcd.print("BPM");
  }

  lcd.print(" |SPO2:");

  if (spo2 < 60) {
    lcd.print("-");
  } else {
    lcd.print(spo2);
    lcd.print("%");
  }

  /* ================ PACKET TRANSMISSION ================ */

  packet.ecg = heartValue;
  packet.ambientTemp = ambientTemp;
  packet.bodyTemp = objectTemp;
  packet.bpm = bpm;
  packet.spo2 = spo2;
  packet.timeSec = seconds;
  packet.iteration = iteration;

  radio.write(&packet, sizeof(packet));

  delay(3000);
}
```

Appendix C : Receiver Unit Code

```
#include <SPI.h>
#include <nRF24L01.h>
#include <RF24.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <MFRC522.h>

/* ================= NRF ================= */
RF24 radio(8, 7);
const byte address[6] = "HLTH1";

/* ================= LCD ================= */
LiquidCrystal_I2C lcd(0x20, 20, 4);

/* ================= RFID ================= */
#define SS_PIN 10
#define RST_PIN 9
MFRC522 rfid(SS_PIN, RST_PIN);

// Authorized UID (F2 68 68 4B)-ID Card
byte authorizedUID[4] = {0xF2, 0x68, 0x68, 0x4B};

/* ================= PACKET ============== */
struct HealthPacket {
  int ecg;
  float ambientTemp;
  float bodyTemp;
  int bpm;
  int spo2;
  unsigned long timeSec;
  int iteration;
};

HealthPacket packet;

/* ================= SESSION CONTROL ===== */
bool accessGranted = false;
unsigned long sessionStart = 0;

void setup() {

  Serial.begin(9600); //9600 baud rate
  Wire.begin();
  SPI.begin();
```

```
  lcd.init();
  lcd.backlight();

  rfid.PCD_Init();

  radio.begin();
  radio.openReadingPipe(0, address);
  radio.startListening();

  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("---Health Monitor---");
  lcd.setCursor(0,1);
  lcd.print("Receiver Unit");
  lcd.setCursor(0,2);
  lcd.print("System");
  lcd.setCursor(0,3);
  lcd.print("Initialising...");
  delay(3000);
}

void loop() {

  /* ================= RFID LOGIN MODE ================= */
  while (!accessGranted) {

    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("Scan RFID card");

    while (!rfid.PICC_IsNewCardPresent() || !rfid.PICC_ReadCardSerial()) {
      // waiting for card
    }

    bool match = true;
    for (byte i = 0; i < 4; i++) {
      if (rfid.uid.uidByte[i] != authorizedUID[i]) {
        match = false;
        break;
      }
    }

    lcd.clear();

    if (match) {
      lcd.setCursor(0,0);
      lcd.print("Access Granted");
      Serial.println("Access Granted");
```

```
      accessGranted = true;
      sessionStart = millis();   // Start 20s timer
      delay(2000);
    }
    else {
      lcd.setCursor(0,0);
      lcd.print("Access Denied");
      Serial.println("Access Denied");
      delay(2000);
    }

    rfid.PICC_HaltA();
  }

  /* ================ SESSION ACTIVE ================ */
  if (accessGranted) {

    // Check if 20 seconds passed
    if (millis() - sessionStart >= 20000) {

      accessGranted = false;   // Reset access
      lcd.clear();
      lcd.setCursor(0,0);
      lcd.print("Session Expired");
      delay(2000);
      return;   // Go back to RFID mode
    }

    if (radio.available()) {

      radio.read(&packet, sizeof(packet));

      Serial.println("---------------------------");
      Serial.println("Receiver Unit");
      Serial.print("ECG Value: ");
      Serial.println(packet.ecg);

      lcd.clear();

      lcd.setCursor(0, 0);
      lcd.print("DR TE:");
      lcd.print(packet.timeSec);
      lcd.print("s Itr:");
      lcd.print(packet.iteration);

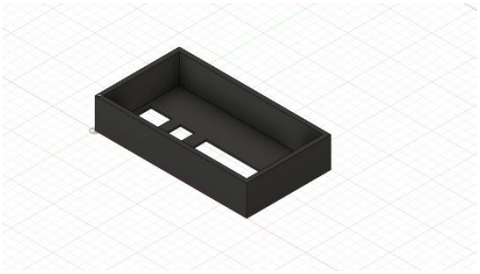
      lcd.setCursor(0, 1);
  lcd.print("ECG Status:");
```

```
    if (packet.ecg > 0) {
      lcd.print("Okay");
    } else {
      lcd.print("Fault");
    }

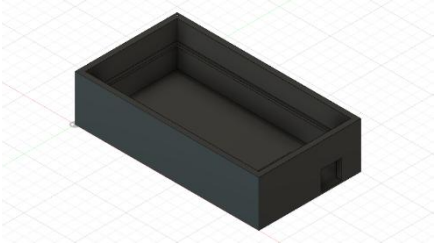
        lcd.setCursor(0, 2);
        lcd.print("AT:");
        lcd.print(packet.ambientTemp,1);
        lcd.print("C |BT:");
        lcd.print(packet.bodyTemp,1);
        lcd.print("C");

        lcd.setCursor(0, 3);
        lcd.print("HR:");
        lcd.print(packet.bpm);
        lcd.print(" |SPO2:");
        lcd.print(packet.spo2);
        lcd.print("%");
      }
    }
  }
```

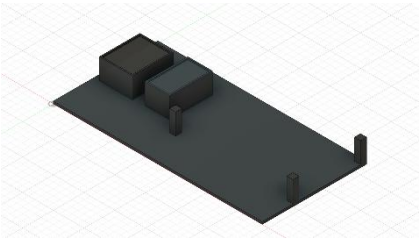
Appendix D : Transmitter Unit 3D Model and PCB Design



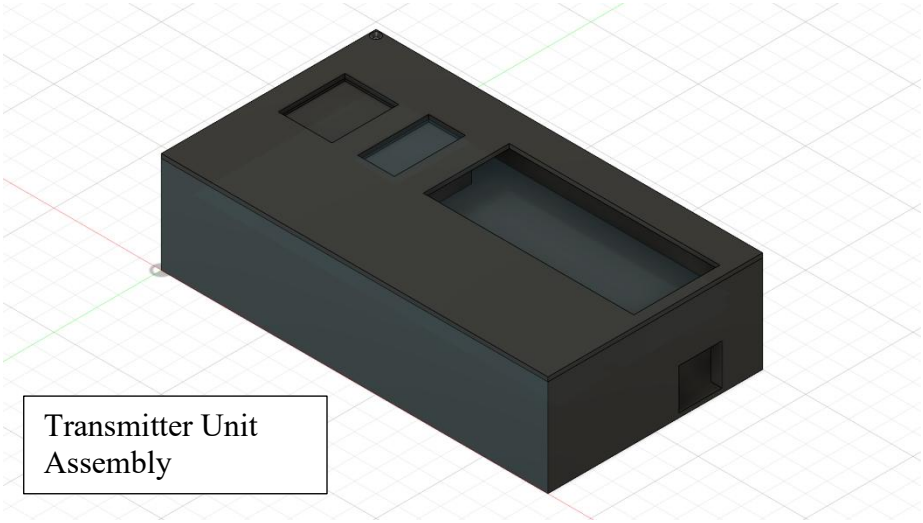
Transmitter Unit Lid



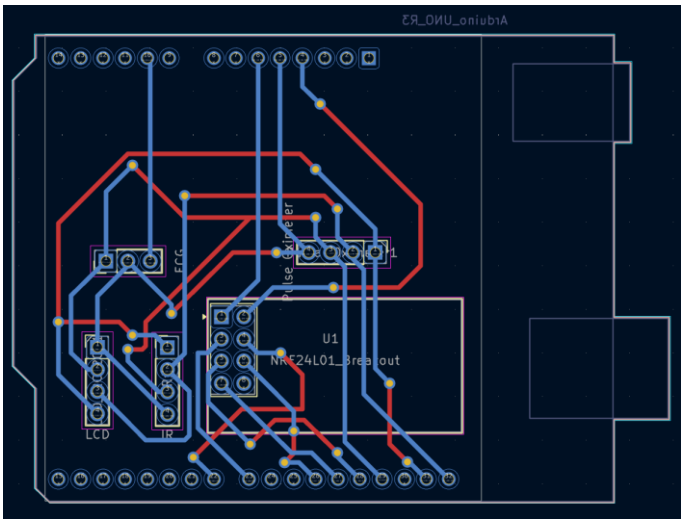
Transmitter Unit Base



Transmitter Unit Slit



Transmitter Unit Assembly

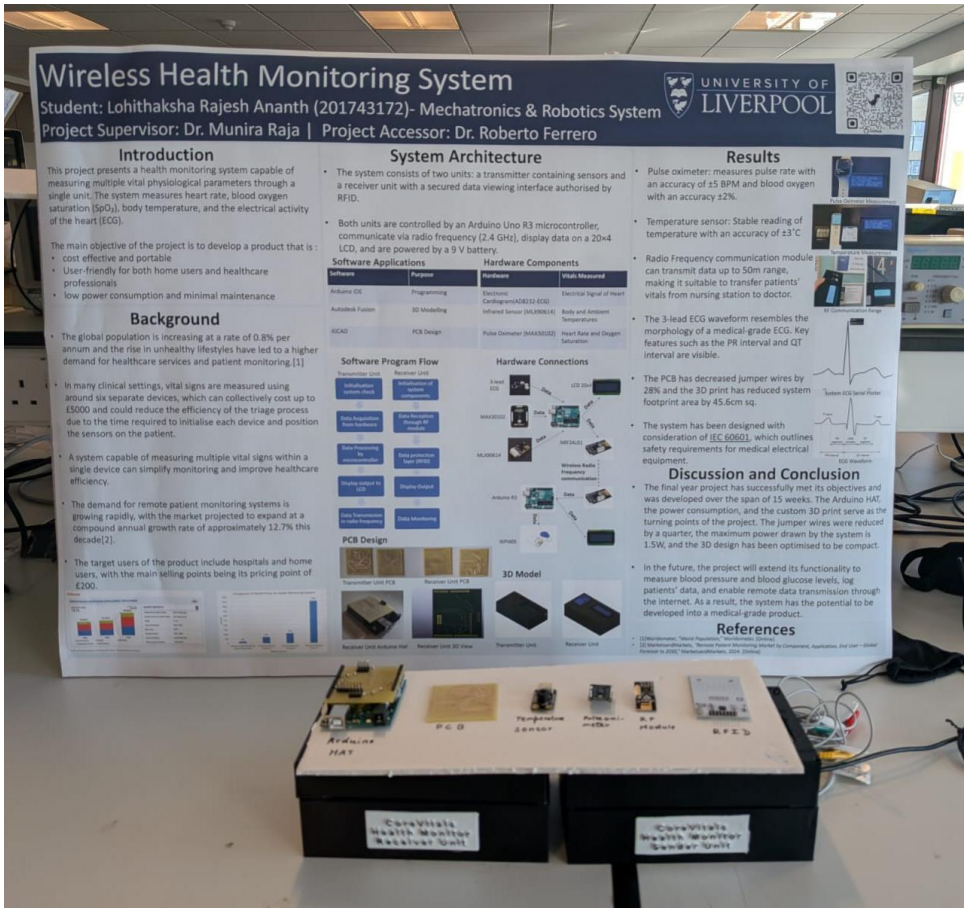


Transmitter Unit PCB Editor View

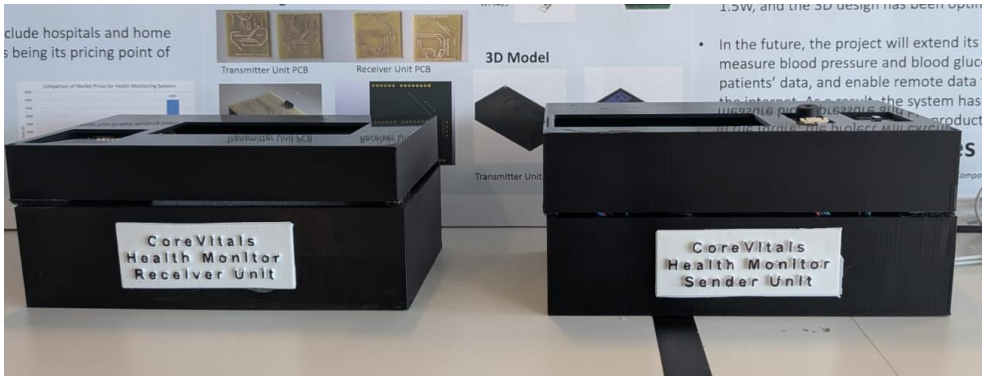
Appendix E : Bill of Materials

Item No.	Component	Quantity	Cost (£)
1	Arduino Uno R3 Development Board	2	University Provided
2	I2C Non-contact IR Temperature Sensor For Arduino (MLX90614-DCC)	1	17.8
3	I2C 20x4 Arduino LCD Display Module	1	13.5
4	Analog Heart Rate Monitor Sensor (ECG) For Arduino	1	19.2
5	MAX30102 Heart Rate and Oximeter Sensor	1	21.1
6	RFID Read and Write Module for Arduino -WPI405	1	8.95
7	nRF24L01 2.4GHz Wireless Radio Transceiver Module	2	2.98
8	X-Power 9V 6LR61 Alkaline Battery	2	9
9	9V battery holder with switch & 5.5mm/2.1mm plug	2	6.8
10	Miscellaneous (3D Print, PCB Design, Solder, Box)	N/A	75 (Estimate)
	Total		174.33

Appendix F : System Photos







Appendix G : Ethical Approval Document

UNIVERSITY OF LIVERPOOL

DEPARTMENT OF ELECTRICAL ENGINEERING AND ELECTRONICS

Application for ethical approval for projects in which the researcher/s and the subject are the same person/people

Approval procedure:

1. Student completes the document *in consultation with the project supervisor*
2. The completed document should be sent to the Department Ethics Lead, Dr Mark Bowden (mbowden@liverpool.ac.uk) for approval
    - If revisions are required, the student should make those revisions, *in consultation with the project supervisor*
3. After approval, the completed form should be kept by the student and included as an appendix in the final report

Please note that, if approval Is granted, that approval applies to the draft plans presented in this document. The project supervisor is responsible for checking final plans and documents (see Section 6.1)

Part 1: Project information

Module:	ELEC 340
Student name:	Lohithaksha Rajesh Ananth
Student ID:	201743172
Supervisor name:	Dr. Raja Munira
Project title:	Remote Health Monitoring System for Patients

Part 2: Project description, including aims

You should describe your project briefly, explaining the reason for the research and the main aims. Take care to describe your project so that it can be understand by a non-expert reader. Your application will be returned if the description cannot be understood.

(Recommended length: 100-200 words)

I am developing a wireless health monitoring system that measures a patient's pulse rate, blood oxygen level, body temperature, and potentially ECG signals. The system has two parts: a sender unit and a receiver unit. The sender unit collects the vital signs using sensors, processes the data, and transmits it wirelessly to the receiver unit. The receiver then displays the information, but access is restricted through a security feature to ensure that only authorised users can view the data. In addition, the sender unit includes a GSM module that automatically alerts a doctor if any vital signs fall outside the normal range.

I chose this project because chronic diseases are becoming increasingly common due to modern lifestyle habits. Patients with long-term conditions often require frequent monitoring, which can lead to overcrowding in hospitals and inconvenience for both patients and healthcare providers. This system aims to support remote health monitoring so that patients can be observed from home, reducing unnecessary hospital visits while still ensuring timely medical attention.

The main aims of this project are:

-To measure vital signs with at least 80% accuracy compared to state-of-the-art medical devices.

-To provide a cost-effective solution that offers a practical balance between accuracy and affordability.

-To offer a user-friendly testing system that can be operated easily without any training.

Part 3: Description of the research activities

You should briefly describe the research activities, especially the activities in which you will self-test or use your own data.

(Recommended length: 200 words)

I will test the pulse oximeter, body temperature sensor, and the ECG module as part of the system development and validation process. During the early stages, I will assess each sensor individually to identify defects and confirm basic functionality. Once the full system is assembled, I will carry out structured self-testing by taking measurements at different times and under varying daily conditions. All data collected will be my own and used solely to evaluate the system's performance and accuracy.

Specific Testing Activities

To understand how my heart rate, blood oxygen level, body temperature, and ECG readings vary across everyday situations, I will test the system under several controlled scenarios:

-After climbing stairs: I will take stairs in the EEE building and immediately measure changes in heart rate, ECG patterns, body temperature, and blood oxygen levels.

-During uphill walking: After a 15-minute uphill walk from accommodation to the University, I will record all four vitals to observe cardiovascular and thermal responses.

-During focused academic work: After working continuously for two hours in the lab, I will measure how sustained concentration affects my vitals.

-During emotional response: I will record my vital signs when receiving academic grades to observe exciting state changes.

-After meals: I will measure heart rate, body temperature, blood oxygen, and ECG changes following food intake.

All collected readings will be logged, compared potentially against a benchmark medical-grade sensor, and used to quantify system accuracy for the final report.

Part 4: Mitigation strategy

The text in this section should describe the strategy that you will adopt to ensure that the risk to the researcher is reduced. In most cases, this requires you and your supervisor to agree strict limits on the types of texting and/or data.

If you are not sure what to write here, you should refer back to the training session slides.

Mitigation Strategy

-Climbing stairs: I will limit stair climbing to only one floor to reduce excessive physical strain and avoid unnecessary spikes in heart rate.

-Uphill walking: I will limit the walking intensity and restrict the duration to a maximum of 15 minutes to ensure the activity remains safe and controlled.

-Emotional response: I will not place myself in stressful or emotionally difficult situations for testing purposes, as this could create unnecessary psychological risk.

-Supervision: My supervisor will receive a video recording of each activity prior to data collection or will be present in person during testing to ensure that all activities remain within the agreed safety limits.

Part 5: Declarations:

You must agree with the statements in this section. If you do not agree with these statements, you can continue with other aspects of your project, but you cannot carry out any activities that require research ethics approval.

You should indicate your agreement by putting a circle around the word "Yes".

5.*1: I understand that my research activities must fall within the restrictions explained to me at the Research Ethics Training Session.*

Yes

5.2: I understand that detailed plans for my activities must be approved by my supervisor before I start.

Yes

Part 6: Approvals

6.1 Supervisor approval:

I approve the draft project plans outline above, and confirm that I will be responsible for checking the detailed plans testing begins.

Name: Dr. Raja Munira

Signature: Mraja Date: 18/11/2025

6.2 Department Ethics Lead:

I approve the draft project plans outline above.

Name: Dr. Mark Bowden

Signature: MBowden Date: 30/11/2025

APPROVED: 30/11/2025

Document version 1.1: Summer 2024

Appendix H : Risk Assessment Document

UNIVERSITY OF LIVERPOOL

Undergraduate Project

Risk Assessment Report

Risk Assessment ID: sglrajes_2025-10-11-14-20-43

Expiry Date: 11/10/2026 *(unless environment has changed)*

Risk Assessor Details:

MWS Username: sglrajes

Name: Lohithaksha Rajesh Ananth

Email: sglrajes@liverpool.ac.uk

Location Details:

Location: Final Year Laboratory

School/Department: Electrical Engineering and Electronics

Building: Electrical Engineering and Electronics, A-Block

Project Title and Project Description:

Year of study: 3

Project Title: Remote Health Monitoring System

Description of work: This project aims to monitor the health of the person by sensing the vital parameters of his/her body and the information can be sent to the receiver unit through RF communication, to be checked by the doctor. This system can be used at hospitals where the health of the patient can be sensed and the information about all the parameters can be transmitted to the receiver unit at the doctor's office using RF communication for timely treatment.

Supervisor Details:

Supervisor Name: Dr. Munira Raja (mraja)

Supervisor Email: mraja@liverpool.ac.uk

Additional Notes:

Key Tables - Risk Score and Action:

Colour Key:

Likelihood						
	5	5	10	15	20	25
	4	4	8	12	16	20
	3	3	6	9	12	15
	2	2	4	6	8	10
	1	1	2	3	4	5
		1	2	3	4	5
		Consequence				

Likelihood/Consequence:

Likelihood		Consequence
Very unlikely	1	Insignificant – no injury
Unlikely	2	Minor – minor injuries needing first aid
Fairly likely	3	Moderate – up to seven days absence
Likely	4	Major – more than seven days absence; major injury
Very likely	5	Catastrophic – death; multiple serious injury

Likelihood(L) x Consequence(C) = Risk(R)

Risk/Action:

Risk	Action to be taken
1-4 Acceptable	No further action but ensure controls are maintained
5-9 Adequate	Look to improve at next review.
10-16 Tolerable	Stop activity and review current controls
17-25 Unacceptable	Stop activity with immediate effect, and rethink the entire experiment

Category:

Category 2 – Projects based on "home-friendly" equipment:

Projects requiring small pieces of equipment that do not require specific health and safety considerations and students can safely use at home (Raspberry Pi's, Arduinos and other similar low-voltage boards with double insulated power supplies).

Voltage and Current:

Any power supplies that can generate current and voltages of **>10mA AND >20V** respectively, can be regarded as potentially extremely hazardous:

Voltage: 12V

Current: 8mA

Identified Hazards and Current Controls:

Likelihood(L) x Consequence(C) = Risk(R)

Identified Hazard	Who can be Harmed	Current Controls	L	C	R
Repetitive action	Myself (Lohithaksha)	-Pre-plan the task and follow safety measures. -Take breaks. -Stay away from distractions -Maintain good posture	3	2	6
Potential for human error	Myself (Lohithaksha)	-Suggesting other lab users to not touch any other persons' project work without permission. -Taking regular breaks to stay alert and maintain accuracy. -Keeping only required components on the table. -Simulating tasks before physically proceeding with the task	2	3	6
Lack of experience	Myself (Lohithaksha)	-Gaining the right skills and knowing procedures to handle lab equipments. -Attending lab related lecture and keeping upto date on lecture learnings. -Consulting professors when in doubt.	2	1	2
Slips, trips, falls	Myself (Lohithaksha)	-Ensuring work place is neat and organised. -Wire management	2	2	4
Electrical equipment	Myself (Lohithaksha)	-Following the right procedures to handle electrical equipment and following the safety protocols specific to the electrical equipment. -Ensuring cables do not tangle. -Switching off devices not in use -Not using power supply with high voltages and currents for long time. -Using right power supply for the specific equipment to avoid unnecessary heating.	2	3	6
Long Hours/Stress	Myself (Lohithaksha)	-Take regular breaks.	2	2	4
Soldering	Myself (Lohithaksha)	-Use fume extraction to safely remove soldering fumes from work area. -Wear safety gloves. -Always place the solder iron on the stand.	2	3	6

Risk Assessor Notes:

Supervisor Notes:
Some risk have high scores, control measures need to be taken seriously

Risk Assessment Status:	Approved by Supervisor.

End of Report
